

Mid-Term Report FYP-1

SkillSeeker

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Executive Summary

This report includes vision of the product that provides the information regarding the scope, purpose and market impact of the product. SRS, which explains the overall functionality of product and external interface requirements. Furthermore, the Use cases explain scenarios of the product for the user and developer. Use case diagram illustrates the use cases in visual format. Expanded use cases describe the use cases in detail for clear comprehension. Domain model provides a conceptual model of product that incorporates both behavior and data. System Sequence diagrams are visual summary of individual use cases. Operational contracts identify system changes when an operation happens. Class diagram shows the logical structure of the system. Architecture diagram represents the abstract overall outline of the software system and the relationships, constraints, and boundaries between components. Data flow diagrams show the flow of data in the system between different external entities and internal system processes. Component diagrams show the relationships that exist between each individual system component.

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Chapter 1: An Overview

1. Introduction

The purpose of this report is to present a detailed plan on the entire development cycle for SkillSeeker. Here we discuss constraint, process flows, delivery and all other aspects of our project.

Finding professional services can be a daunting task for layman users. The process often takes upwards of 60 minutes, involving extensive searching to identify valid professionals. Users frequently face the frustration of making countless calls to multiple service providers, only to be met with complex decision-making processes that require more expertise than they possess. Moreover, the lack of specific niche search options further complicates the task, leading to a significant amount of unnecessary legwork. This inefficiency and the time-consuming nature of the search process hinder users from quickly and easily accessing the professional services they need.

We propose a system for Layman users to find these professional services through the use of extremely simple prompts.

1.1 Problem Statement

Let's take a real-world example, Ahmed has been experiencing persistent back pain and needs to find a doctor. However, as he searches online, he becomes overwhelmed by complex medical terms like "orthopedic specialist" and "rheumatologist." Unsure of what he needs, John spends nearly an hour scrolling through websites and making calls to clinics. Each conversation adds to his confusion, as receptionists use technical language he doesn't understand. What should have been a simple task turns into a frustrating and time-consuming ordeal, leaving John no closer to finding the right doctor.

Layman users often struggle to locate the right professionals, particularly in specialized fields like law, computer science, and medicine. For instance, someone needing legal advice for a business contract may not know whether to consult a corporate lawyer or a contract attorney, while a person facing a cybersecurity issue might be unsure if they need a cybersecurity expert or a network engineer. In the medical field, a user with back pain might be confused about whether to see an orthopedic specialist, rheumatologist, or chiropractor. This confusion leads to wasted time and resources, and often results in users receiving help that isn't tailored to their specific needs, further complicating their situations.

1.2 Problem Solution

Introducing an all-in-one LLM application designed to bridge the gap between layman users and professional services. This chatbot-driven solution allows users to input their needs in simple terms, without requiring any prior knowledge of specific professions. The application then generates a comprehensive list of tailored professionals, complete with user customization options and role management features. Additionally, it offers seamless booking functionalities, ensuring that users can easily connect with the right service providers.

Users can provide specific directions or cases for their searches, such as requesting a lawyer with experience in intellectual property who has experience in retail sector disputes or a doctor

specializing in pediatric neurology with a keen interest in psychotherapy and has dabbled in surgeries. The chatbot will process these criteria and generate a final output that showcases professionals with the closest match to those needs. The results will highlight individuals' expertise, proximity, and ability to address the specified problems, ensuring users connect with the most qualified service providers for their unique situations.

1.3 Project Vision

1.3.1 Overview

In summary, this project aims to revolutionize how users access professional services by introducing an all-in-one LLM application that simplifies and personalizes the search process. By leveraging a fully automated chatbot, users can easily input their needs and receive tailored recommendations. The application includes comprehensive features such as multilingual support, voice interaction, and an integrated booking system. With additional functionalities like a "Near Me" feature and a sophisticated history and personalization module, users will benefit from a streamlined, efficient, and user-friendly experience, ensuring they find the right professionals quickly and effectively.

1.3.2 Definition, Acronyms or Abbreviations

- LLM - Large Language Model
- Vector Embeddings – Database System for Retrieval
- AI - Artificial Intelligence
- MoE/MoA – Mixture of Experts/Mixture of Agents
- CrewAI – Agentic Approach

1.4 Positioning

SkillSeeker targets layman users that have little or no understanding of professions, allowing them to get paired up with individuals that are more suited to their needs as a whole

Users can provide specific directions or cases for their searches, such as requesting a lawyer with experience in intellectual property who has experience in retail sector disputes or a doctor specializing in pediatric neurology with a keen interest in psychotherapy and has dabbled in surgeries

1.4.1 Problem Statement

The problem of	Locating Professionals
affects	All Layman users and some other agencies/users
the impact of which is	Difficulty in conveying requirements
a successful solution would be	An all-in-one LLM application designed to bridge the gap between layman users and professional services. This chatbot-driven solution allows users to input their needs in simple terms, without requiring any prior knowledge of specific professions.

Table 2: Problem Statement

1.4.2 Product Position Statement

For	Users of all types, mostly Layman
Who	Cannot convey functionality and requirements
SkillSeeker	Is a software product
That	Runs on web browsers and allows users to locate professionals that fit their criteria.
Unlike	Available systems that are too complicated and do not allow basic layman level inputs and specific criteria checks
Our Product	Enhances the soft skills of employees at organizations of any scale. It quantifies their soft skills using different metrics extracted from audio and video recordings of them communicating in real life based settings, and then provides recommendations for improvement based on the results.

Table 3: Product Position Statement

1.5 Scope

1.5.1 Overview

1. **User Login:** Upon logging in, users are greeted with their personalized preferences if they are returning users. This customization ensures a tailored experience right from the start.
2. **Specify Needs:** Users can either describe their problem and the desired solution or provide extremely specific criteria for the services they need. The chatbot will use this information to generate a list of relevant professionals.
3. **List-Based Results:** The application presents a list of professionals based on the user's criteria. Each entry provides details about how well they match the user's needs.
4. **Communication with Chatbot:** Users can then interact with the chatbot to discuss the professionals on their list. They can ask questions about each individual's expertise, specific details, and how they can address their problems. This step helps create a comprehensive persona of the professional and assesses their suitability.
5. **Final Decision:** After evaluating the options and discussing them with the chatbot, users can make a decision on which professional to contact.
6. **Appointment Booking:** Users can request the chatbot to send a reminder for booking the appointment. The system will send an appointment confirmation via email or SMS, ensuring users are notified and reminded of their upcoming engagement with the chosen professional.
7. **Professions:** Currently, we plan to embed these professions:
 - All Major CS Fields (Especially the ones present in FAST)
 - Major Medical Fields
 - Major Lawyer Fields
 - Notable Marketing Fields
 - SEO/Journalism Fields
 - Many Service-Based Locations (Sports related, entertainment related)

1.5.2 Target Audiences

- **Layman Users:** Individuals with little understanding of professional categories who need guidance in finding the right service.
- **Newcomers:** Users who are new to their surroundings or locationally challenged, requiring assistance in navigating local professional services.
- **Time-Sensitive Users:** Those who need quick, efficient solutions due to time constraints or urgent cases.

1.5.3 Goals

- **Rapid Access:** Provide a comprehensive list of reliable professional services in under a minute.
- **Tailored Searches:** Enable users to conduct highly specific criteria-based searches and customizations.
- **Informed Decision-Making:** Facilitate further communication with the chatbot to discuss and evaluate the proposed professionals.

- **Seamless Booking:** Offer a booking feature to directly connect and communicate with the selected service providers.

1.6 Modules

1.6.1 Prompting Module

This module is responsible for understanding the user prompts and creating context for classification and retrievals. It deconstructs the prompt and creates blocks of usable data essentially

- Gets info from user for future querying
- Find the most suited profession for this task
- Responds with the correctly classified profession

1.6.2 Profession Response

This module is responsible for giving the answers to the user while maintaining the correct flow and understanding of the initial queries. After a conversation chain with the assistant, the bot will respond with the results

- **Conversation Chain**
- **Response/Output**
- **Detailed Info on Users**

1.6.3 Persona Module

This module is our most interesting one, we essentially create a module that works as a person, this person can then give an interview to the user that wants to find a profession best suited to their needs.

- **Conversation Based Interview**
- **Full Q/A Session**
- **Persona Development**

Chapter 2: Project Requirements

2.1 Product Features

A vast description of all major features and requirements of SkillSeeker as a whole.

2.1.1 *Features*

- A memory system that allows the assistant to remember details about the user and their requirements.
- The model that will translate queries and give a more formal tone
- A list-based answer for all users
- Multi-language support
- Differentiation between niches and professionals

2.1.2 *Usability*

- Users can interact with SkillSeeker by communicating in a web-based platform.
- Users can input any requirements for any of the available professions
- Users can then interview the personas for more details

2.1.3 *Reliability*

Our system will provide accurate results as well as an interface that is simple and easy to use. The model itself will be hosted via MERN stack and will contain fallback features that help keep it hosted

2.1.4 *Performance*

Response time and results displayed will be quick, to compete against other forums like ours. The model itself will be mostly API based, allowing for other users to not have to use GPU resources, aiding their resource management and ours

2.1.5 *Supportability*

Extremely scalable system using vector embeddings and other systems. This system of scalability allows us to keep adding professionals without the fear of overburdening our system with unnecessary datasets.

2.2 Constraints

2.2.1 *Implementation Constraints*

- Training our own models and API usage
- A MERN system for UI
- MySQL as backend

2.2.2 *Process Constraints*

Number of professional types and how they differ from others of its niche, forcing us to adapt the model to accommodate items further

2.2.3 *Physical Constraints*

None

2.2.4 *Environmental Constraints*

None (ATM)

2.3 Overall Description

2.3.1 *Product Functions*

The main functionality of SkillSeeker will be

- Locating professionals
- Interviewing personas
- Bookings
- Input translations

2.3.2 *Layman User*

This section describes the characteristics of the Trainee. Trainee should have experience of using desktop-based web browser applications.

- Experience of using a desktop-based web browser.
- Satisfactory level of understanding of distinctive interfacing.
- User should have a complete understanding of the features of product.

2.3.3 *Operating Environment*

For this project, the operating environment must have a stable connection to the internet to access the application on the web.

2.4 *External Interface Requirements*

2.4.1 *User Interfaces*

User interface will be similar to other training applications, where user can first register their account. After logging in, the user can add any requirements or criteria's and based on that receive their answer, and then they can go for interviewing personas.

2.4.2 *Hardware Interfaces*

SkillSeeker is not highly dependent on any hardware interface, it requires only a computer and internet with web browser. However, the web browser should be accessed from a desktop for the optimal user experience.

Chapter 3: The System

3.4 System Features

3.4.1 Login

Description and Priority

Users can login to the application through the UI interface and be greeted with the system. A simple security check as well as DB retrieval will be used to check if the correct person has logged in or not

Stimulus/Response Sequences

1. Get user ID.
2. Get user password.
3. Show previous conversations (If any)

Functional Requirements

REQ-1: The login ID must be valid.

REQ-2: The password must match to corresponding login ID.

3.4.2 Select Professional

The user gives a query and it is classified into a certain profession. The prompt may be extremely simple allowing it to be dissected by our model for easier uses in future manipulation

Stimulus/Response Sequences

1. Enter the assistant tab
2. Give requirements

3. Manage results

3.4.1 Conversation

Description and Priority

User chats with the chatbot about the initial query. A complete conversation chain where the bot creates new questions dynamically based on the answers provided, these answers can help the model find more people useful for the user.

Stimulus/Response Sequences

1. Users receives sends requirements
2. User receives new questions
3. Answers are provided

Functional Requirements

REQ-1: At least one initial prompt with some detail

3.4.2 Manage Results

Description and Priority

Users can see and manage all users that fit the profile and check who fits where. On a side tab, there will be all the professionals best suited, the user can simply click on them and view what works best.

Stimulus/Response Sequences

4. Users receives output
5. User chooses a person
6. User starts the interview

Functional Requirements

REQ-1: At least one full conversation

3.4.3 Booking

Description and Priority

Once decided, a user may book a professional through the app. After choosing a professional, the user can click on his profile and ask the bot to send a booking to him through whatever forum available

Stimulus/Response Sequences

1. User chooses a user
2. User enters his tab (Details)
3. User tells the bot to book details

Functional Requirements

REQ-1: At least 1 chosen user

3.5 Other Nonfunctional Requirements

3.5.1 Performance Requirements

- **Internet Connection:** The Computer/Laptop should be connected to internet.

3.5.2 Safety Requirements

The product is safe as no user data other than the initial convo is stored in the user preferences in database

3.5.3 Software Quality Attributes

Following are the software quality attributes of our system.

3.5.3.1 Efficiency

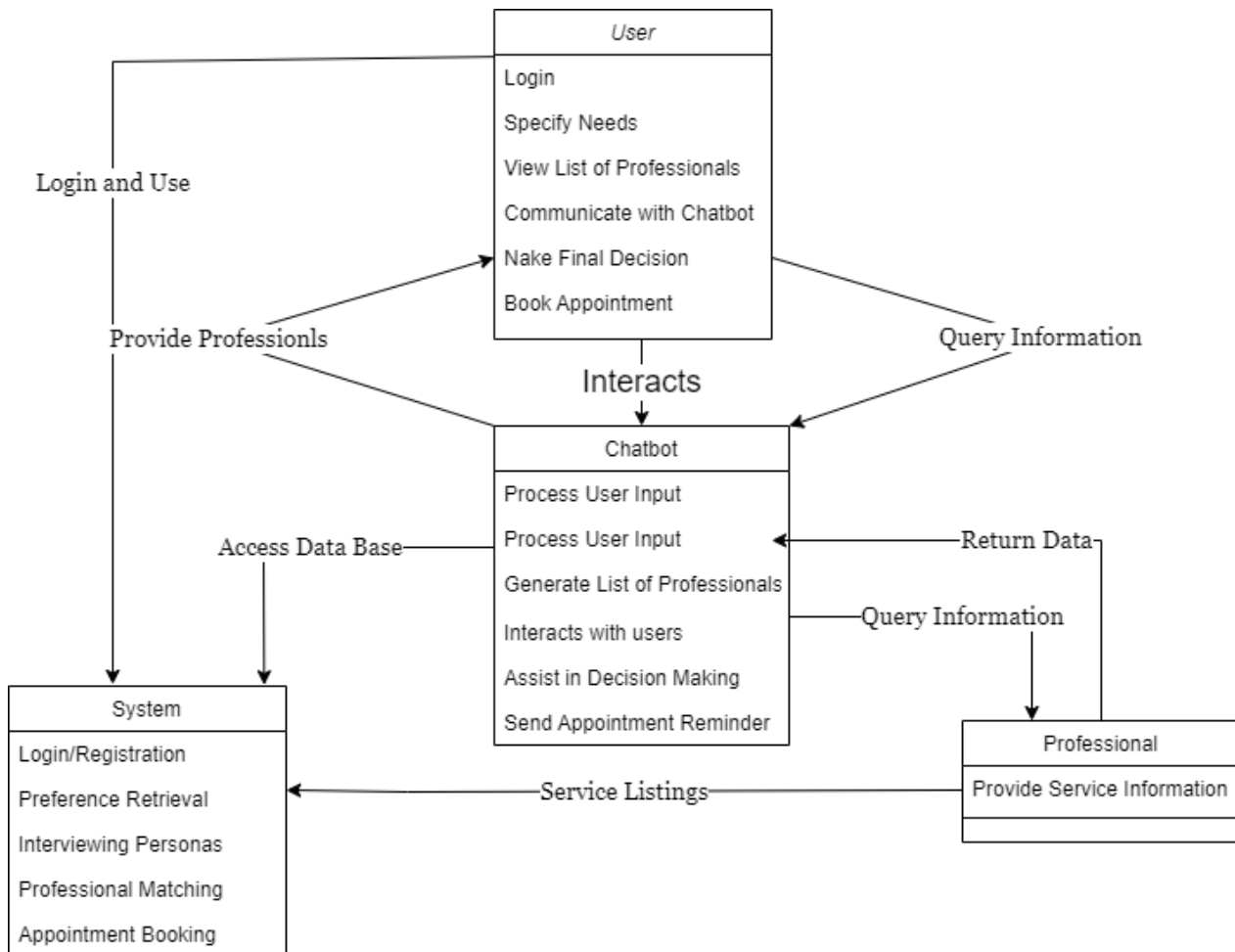
The system analyzes the prompt in detail and gives best users

3.5.3.2 Accessibility

The product is designed for people who don't understand complex criteria's and details.

3.6 Use Case Diagram

Figure: Use Case Diagram



3.7 High Level Use Cases

3.7.1 Login

Use Case	Login
Actor	User
Type	Primary
Description	The system will check if the user has conversed with the bot before and gives recommendations based on that.

3.7.2 Select Professional

Use Case	Select Professional
Actor	User
Type	Primary
Description	User gives an input based on their needs, this can be document, list, or problem based.

3.7.3 Conversation

Use Case	Conversation
Actor	User
Type	Primary

Description	The assistant will ask follow-up questions to get a better idea of how to understand the user problem.
--------------------	--

3.7.4 Manage Results

Use Case	Manage Results
Actor(s)	User
Type	Primary
Description	The user can get individually see each professional, their pros, cons and other details and decide who they want to discuss further

3.7.5 Persona Interview

Use Case	Persona Interview
Actor(s)	User
Type	Primary
Description	The user can get interview a sort of persona of the professional chosen and see how well they align based on their prompts and such.

3.7.8 Booking

Use Case	Booking
Actor(s)	User
Type	Primary

Description	The user can ask the agent to book the user for him
--------------------	---

3.8 Expanded Use Cases

3.8.1 Select Professionals

Use Case	Select Professionals
Actor	User
Level	User Goal
Stakeholders and Interests	User: Wants to get some professionals
Preconditions	• User has logged into the app • He is somewhat clear on what he wants
Post Conditions	• The prompt has been analyzed • Results displayed
Main Success Scenario	This case begins when user logs in 1. The user will provide a prompt, very layman level 2. The prompt is deconstructed 3. The prompt is then classified which profession it belongs to 4. The prompt has been successfully analyzed and the user is then notified through the next steps.
Alternate Scenario	This will ONLY occur if the prompt has not been classified into any available area, meaning the prompt was severely lacking or it is not one we offer
Frequency of Use	High
Technology and Data Variation Lists	The user must have a stable internet connection

3.8.2 Conversation

Use Case	Conversation
Actor	User
Level	User Goal
Stakeholders and Interests	User wants to give more info on his problem
Preconditions	User has given initial prompt
Post Conditions	User gets the results
Main Success Scenario	- The user gets a few questions based on his prompt - These questions are answered - These questions build up confidence in the entire prompt for the retrieval chain - Results are displayed
Alternate Scenario	User does not provide valuable info in the conversation, leading in worse results
Frequency of Use	High
Technology and Data Variation Lists	User must have a stable internet connection; User must also understand his problem to a small degree

3.8.3 Manage Results

Use Case	Manage Results
Actor	User
Level	User Goal
Stakeholders and Interests	User wants to see details on each professional
Preconditions	User has participated in entire convo chain
Post Conditions	User gets the needed results
Main Success Scenario	- The user looks at all the individuals - The user checks to see who he wants to interview - The user makes a choice

Alternate Scenario	User does not choose anyone
Frequency of Use	Normal
Technology and Data Variation Lists	User must know the kind of person he wants

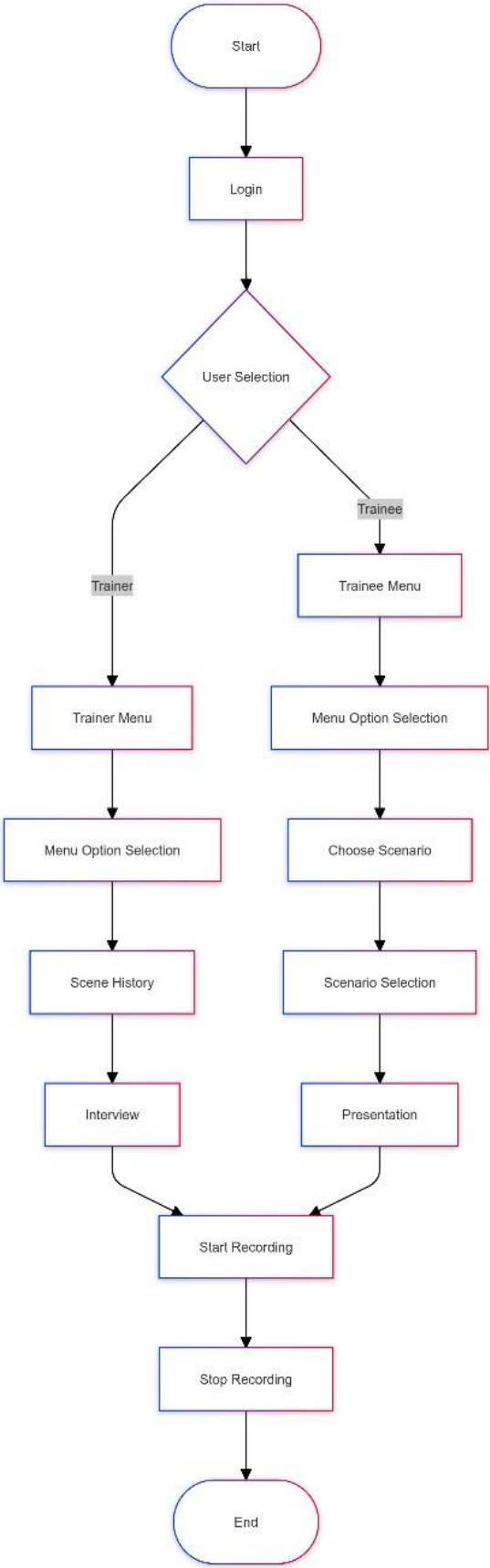
3.8.4 Persona Interview

Use Case	Persona Interview
Actor	User
Level	User Goal
Stakeholders and Interests	User interviews the persona of a user
Preconditions	1. User has chosen a person to interview 2. The assistant receives the new questions 3. The user makes a decision
Post Conditions	The user can tell the bot what he wants to do after choosing the user
Main Success Scenario	1. The user gives his entire interview 2. The user analyzes result 3. The user makes a decision over who he wants
Alternate Scenario	User does not take interview, either leaves or chooses someone directly
Frequency of Use	Normal
Technology and Data Variation Lists	-

3.8.5 Booking

Use Case	Create Interview Session
Actor	User
Level	User Goal
Stakeholders and Interests	User wants to book a user for a certain time, physically or digitally
Preconditions	• User has decided on a professional
Post Conditions	• User gets confirmation on booking
Main Success Scenario	1. User decides on a user 2. User then books on a user 3. The chatbot gives confirmation 4. The booking info is sent to the professional
Alternate Scenario	User does not book user
Frequency of Use	Low
Technology and Data Variation Lists	User must have decided on a user

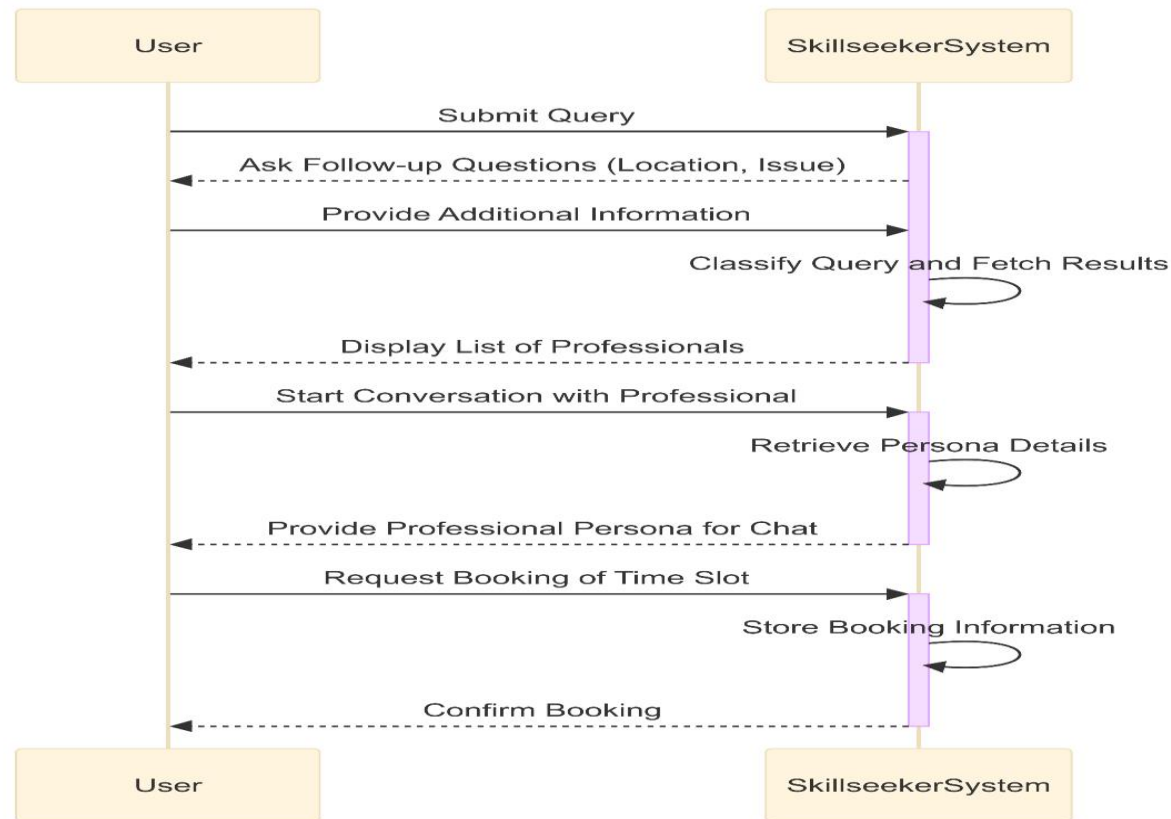
3.9 Activity Diagram



Fig# 2

3.10 System Sequence Diagrams

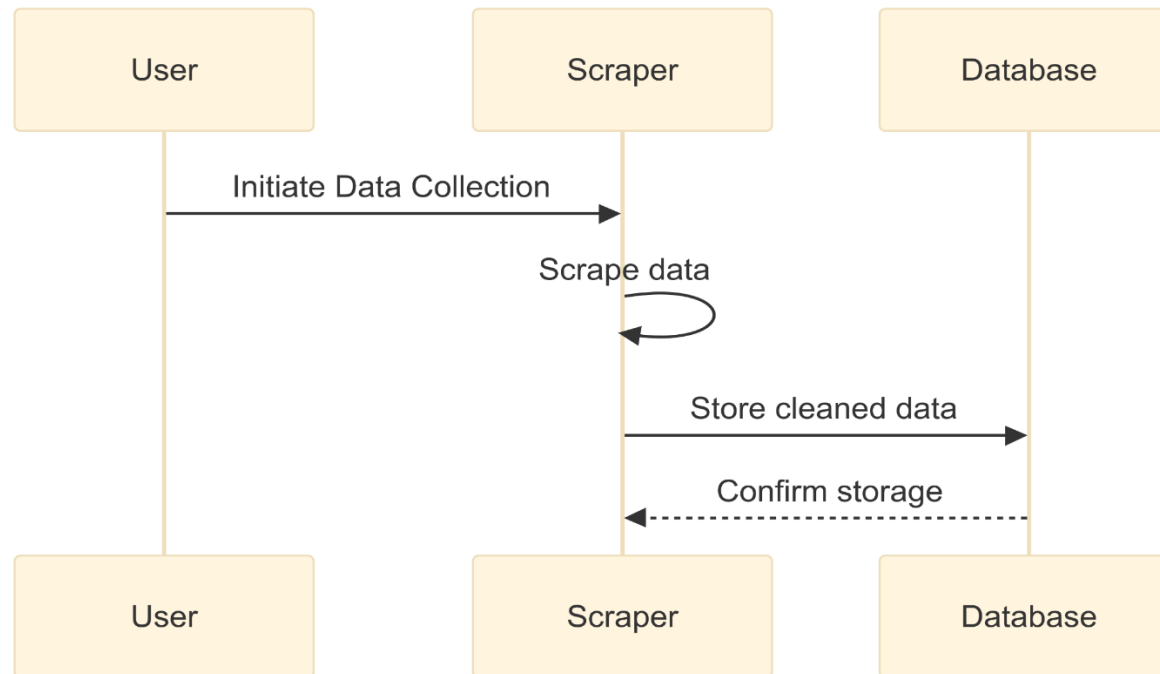
3.10.1 Main Diagram



Fig# 3

Depicts the sequence flow of a query submission by a user within the Skillseeker system, including follow-up, professional listing, chat initiation, and booking confirmation.

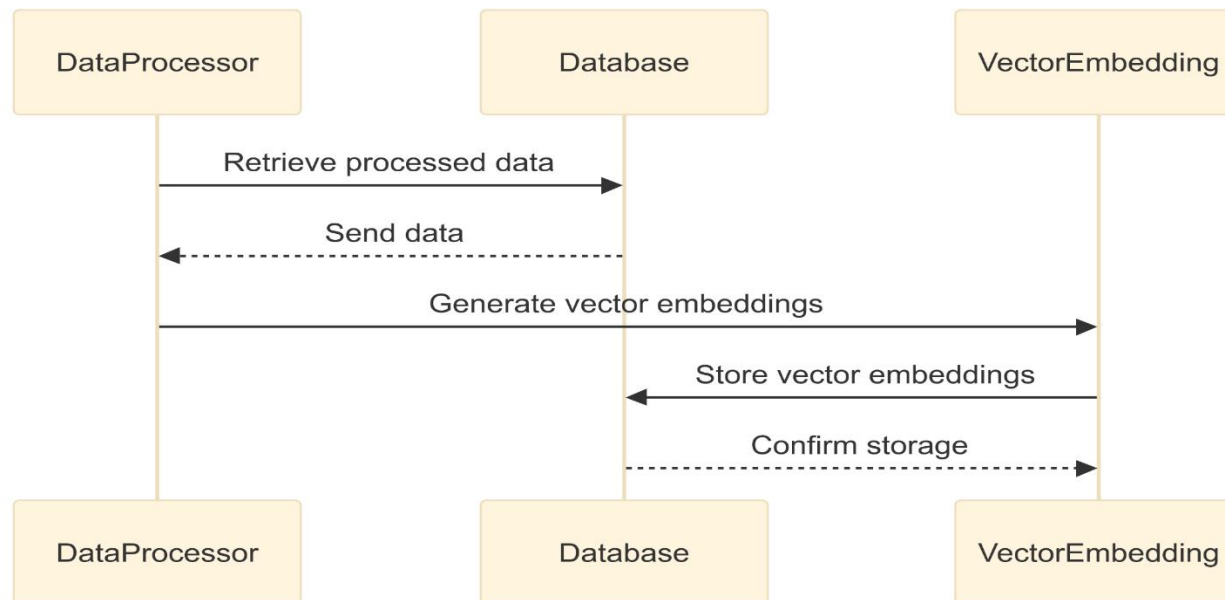
3.10.2 Data Collection



Fig# 4

Illustrates the data collection process using a scraper, from initiating the scrape to storing and confirming data in the database.

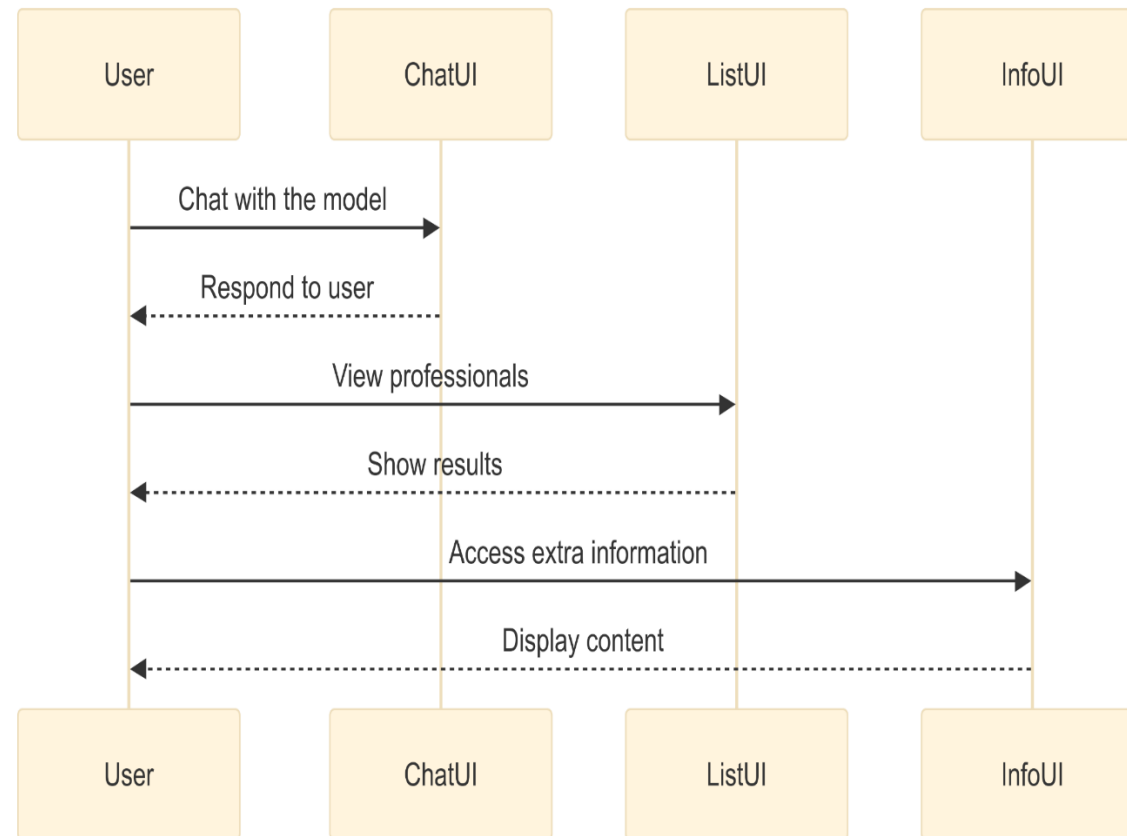
3.10.3 Vector Embeddings



Fig# 5

Represents the sequence of generating vector embeddings, starting from data retrieval from the database to the storage and confirmation of the embeddings.

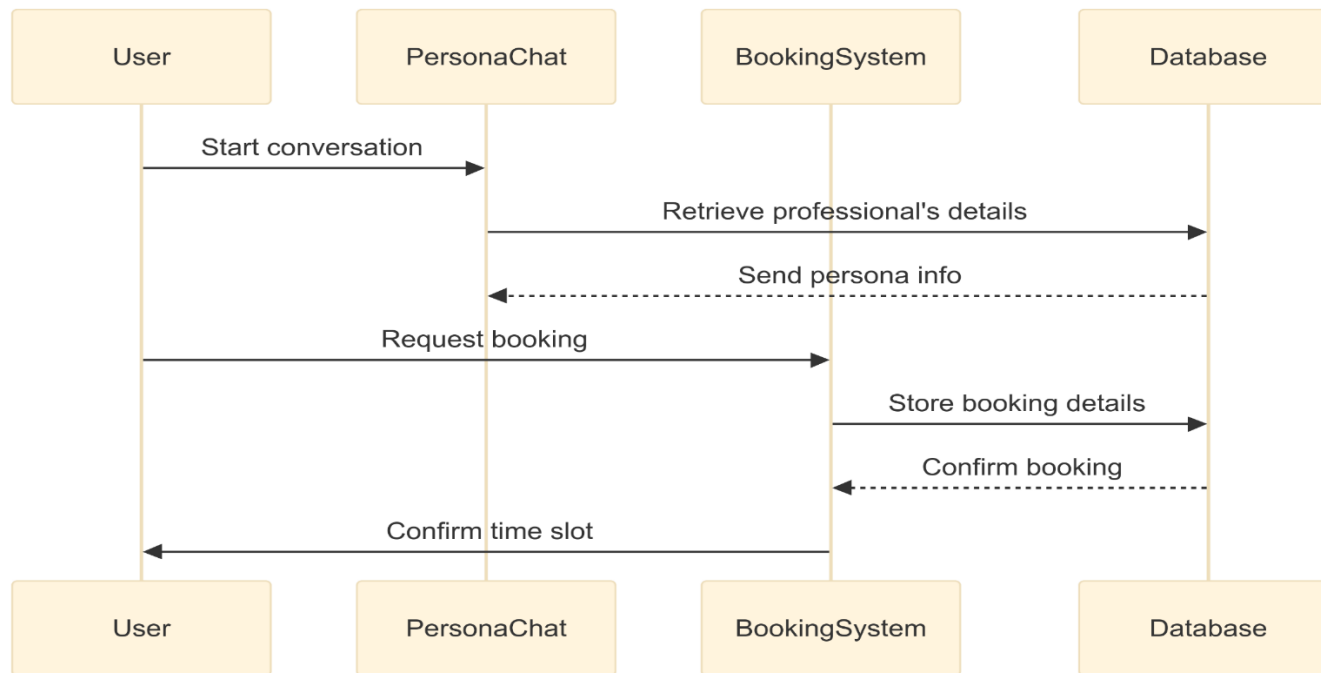
3.10.4 UI Interaction



Fig# 6

Showcases the interaction between the user and different UI components (ChatUI, ListUI, InfoUI) for viewing professionals, chatting, and accessing additional information.

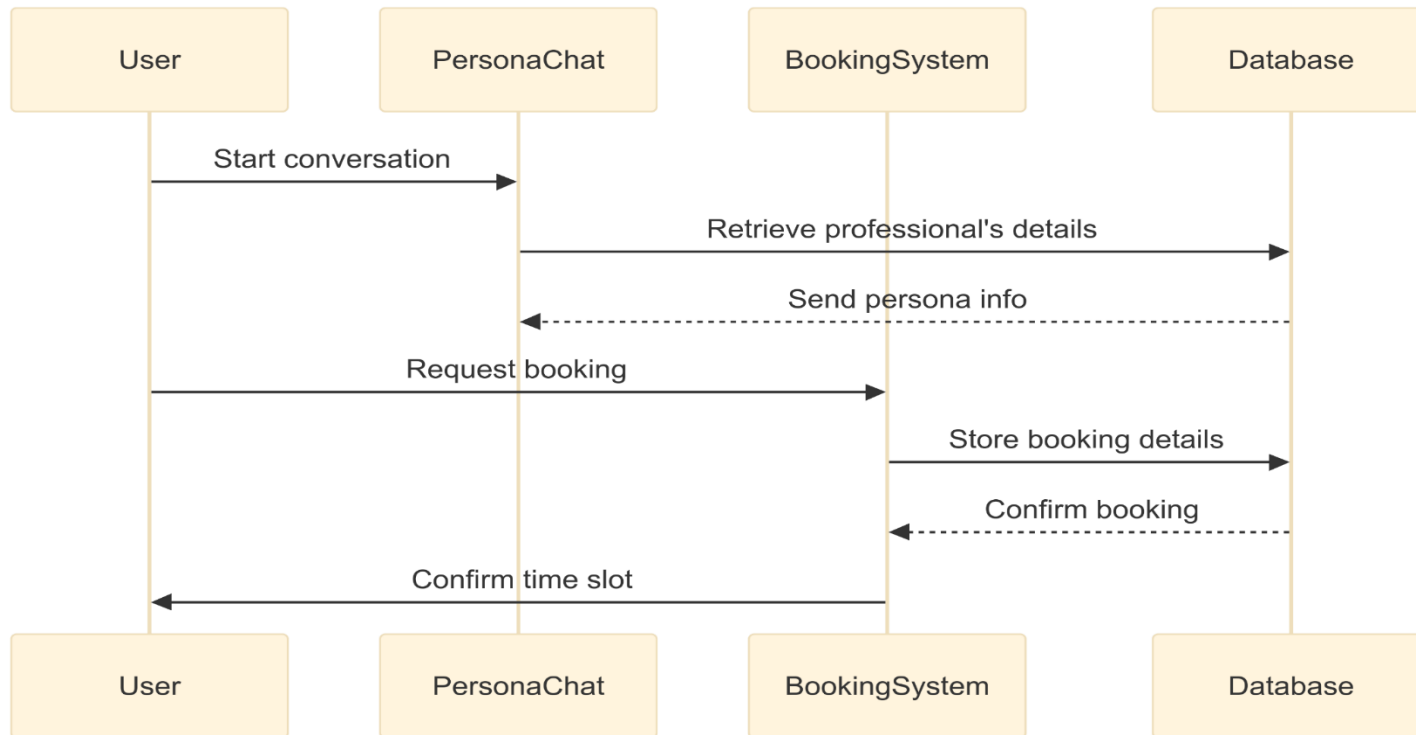
3.10.5 User Interaction with Professionals



Fig# 7

Details the process of a user starting a conversation via PersonaChat, retrieving professional details, requesting a booking, and confirming the time slot through the BookingSystem.

3.10.6 User Query Handling



Fig# 8

3.11 Class Diagram

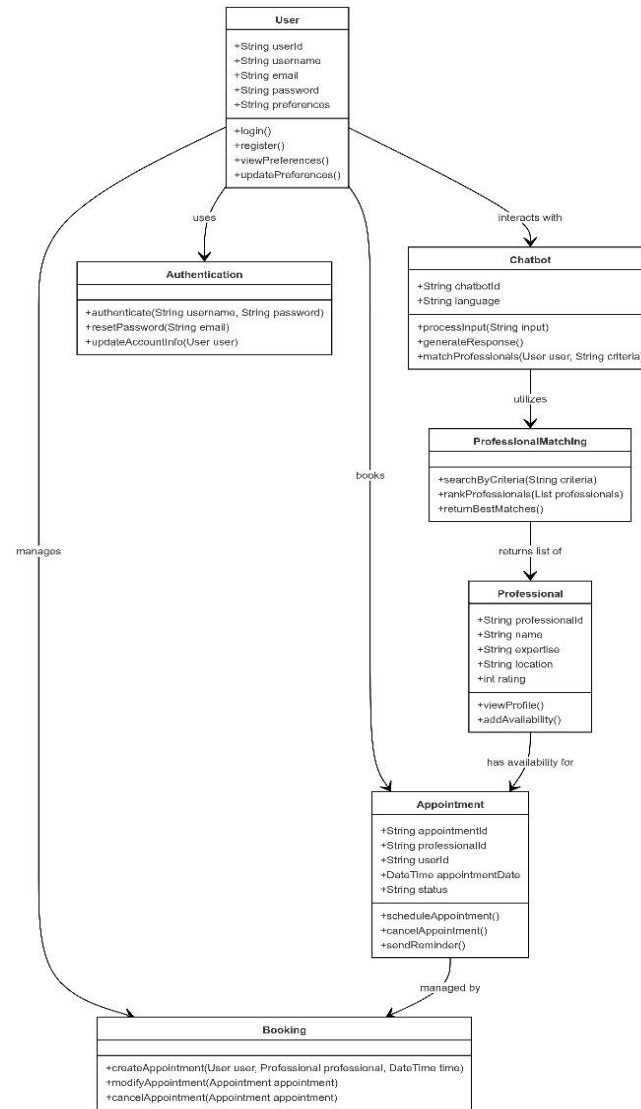
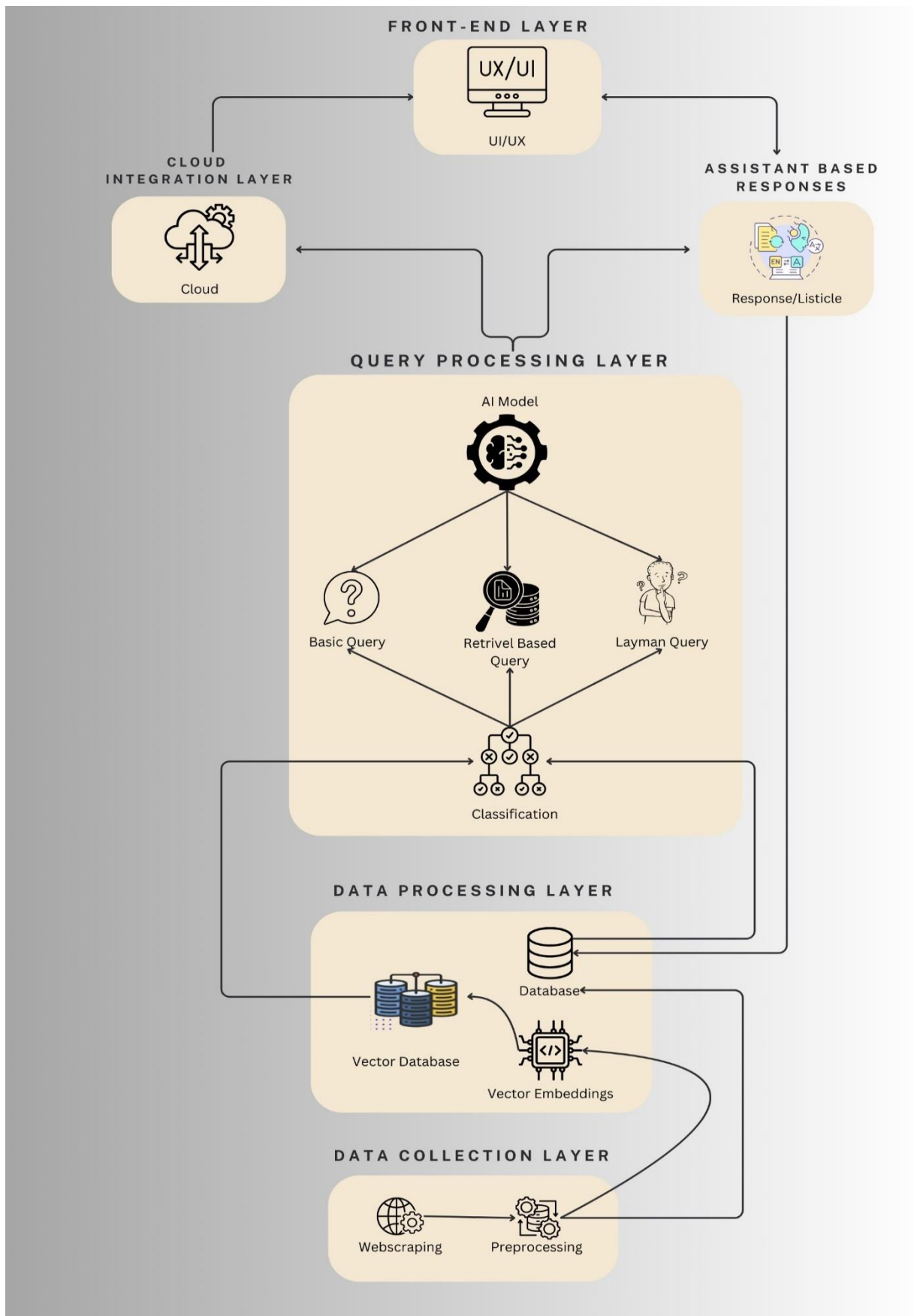


Figure # 9 Class Diagram

3.12 Architecture Diagram



3.13 Data Flow Diagram

3.13.1 Level 0 DFD (Context Diagram)

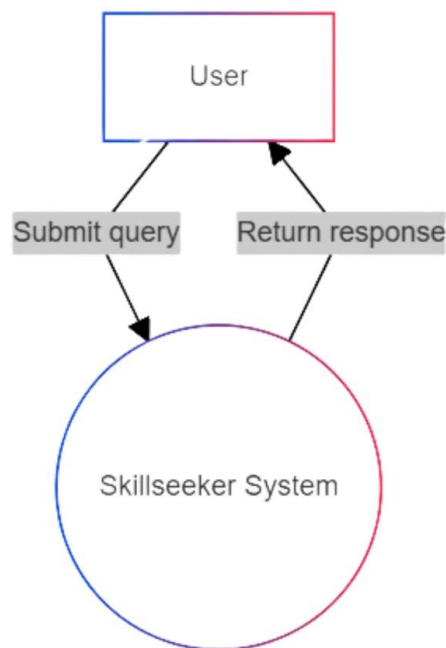
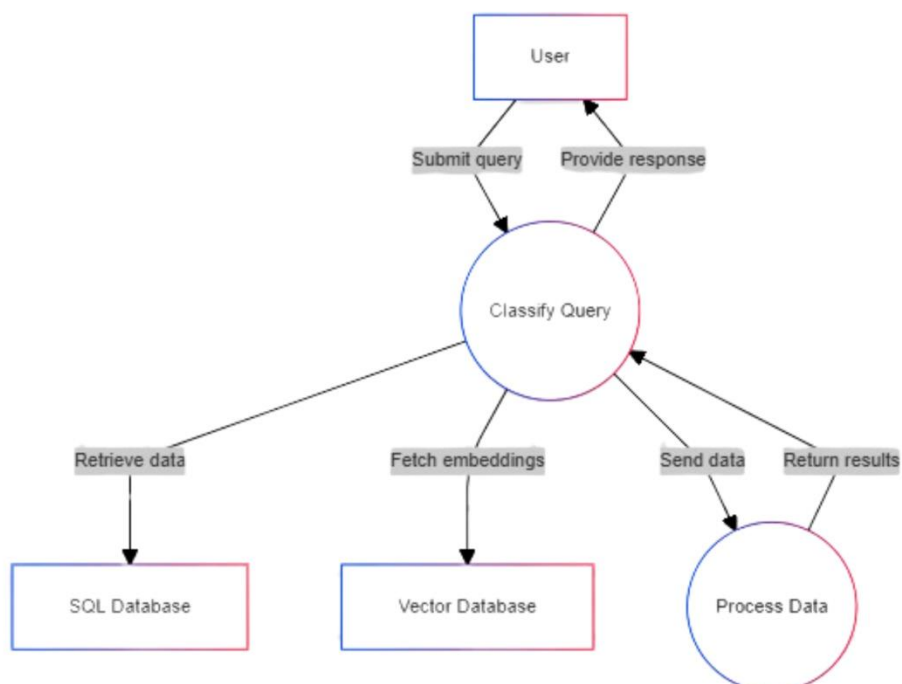


Fig # 11

The diagram represents a Level 0 Data Flow Diagram (DFD) or Context Diagram for the "Skillseeker System." In this system, the "User" submits a query to the "Skillseeker System," which processes the request and returns a response to the user.

3.13.2 Level 1 DFD (Overview of Key Processes)



Fig#12

This diagram depicts the interaction between the user, query classification, SQL database, vector database, and data processing. It shows how the system retrieves and processes data to return a response to the user.

3.13.3 Level 2 DFD - Classifying Query

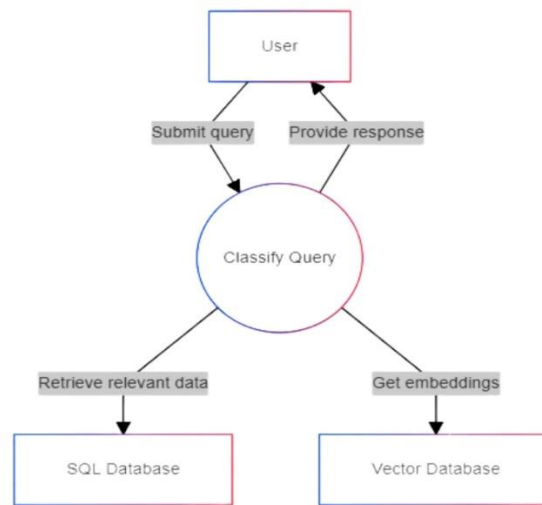
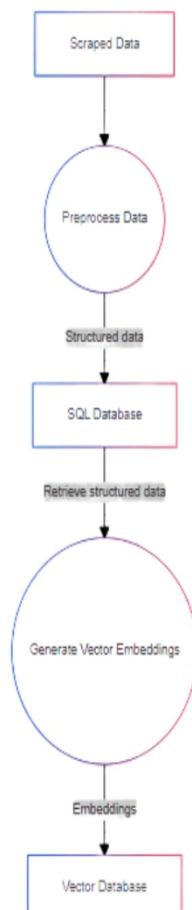


Fig #13

This diagram illustrates the detailed process of query classification, including retrieving relevant data from SQL and vector databases, followed by generating an appropriate response for the user.

3.13.3 Level 2 DFD – Processing Data



Fig#14

This diagram focuses on how data is processed in the system, starting from scraping, preprocessing, storing structured data in an SQL database, and generating vector embeddings for storage in a vector database.

4.1 Conclusion

In summary, this project aims to revolutionize how users access professional services by introducing an all-in-one LLM application that simplifies and personalizes the search process. By leveraging a fully automated chatbot, users can easily input their needs and receive tailored recommendations. The application includes comprehensive features such as multilingual support, voice interaction, and an integrated booking system. With additional functionalities like a "Near Me" feature and a sophisticated history and personalization module, users will benefit from a streamlined, efficient, and user-friendly experience, ensuring they find the right professionals quickly and effectively.

This application will allow for a streamlined process, complete transparency, and up to date results for any profession.

4.2 References

Following are the sites that will serve as a template to our project via user data

<https://www.fiverr.com/>

<https://oladoc.com/>

<https://www.upwork.com/>

<https://www.hg.org/>

<https://clutch.co/>

To better understand our process flows and tools, these documentations can help:

<https://www.crewai.com/>

<https://github.com/meta-llama/llama3>

<https://huggingface.co/blog/mlabonne/agent-ic-datagen>

<https://nexla.com/ai-infrastructure/prompt-tuning-vs-fine-tuning/>